Reminder

1. No late submission is allowed.
2. Please upload your digital version of answer for Assignment 1(PDF file format). All Stata codes should be listed at the end of the file.
3. Some necessary Stata codes for reference:

   - Generate variable

   generate variable=variable1 / variable2

   - Log transformation

   (We add 1 here to avoid generating missing values when the variable value is equal to 0)

   generate log_variable=log(variable+1)

   - Quadratic form

   generate x_square=x^2

   - Run regression

   (where y is the dependent variable name, x is the independent variable name)

   reg y x

   - Compute the fitted value

   (1) Fitted value y (yhat is the variable name of fitted value you name)

   predict yhat

   or manually (β1 and β0 are values that estimated by regression)

   generate yhat=β1*x+β0

   (2) Residual (r is the variable name of residual you name)

   predict r, residuals

   or manually

   generate r=y-yhat

   (3) Sum of residual

   egenerate sum_residual=sum(r)

4. Group discussion is welcomed.
5. Multiple attempts are allowed, and the last attempt will be used for grading.

Problem Set 1
ECO3121 - Fall 2022

September 21, 2022

Due 3PM, 09/30/2022
Please remember to submit your Stata code and requested output as it will be graded.

Question 1

Download from the blackboard site and load into STATA the agricultural household data `aghousehold.dta`. The main data we use is the National Fixed Point Survey (NPFS), which is a nationally representative panel dataset (unbalanced) of roughly 5000 households in 88 villages between 1995 and 2002. It is collected by the Ministry of Agriculture and Rural Affairs of China.

Since this will be a univariate model, we'll be estimating regressions of the form:

$$yield_i = \alpha + \beta_1 fertilizer_i + \mu_i$$

First, let's analyze the effect of fertilizer application on yield.

1. Since we are learning simple linear regression and multiple linear regression at current stage, you can limit the sample to households in year 2002 by running code "keep if year==2002" in Stata. you can generate variable yield (output per unit of land) via $\frac{hx95_nh125}{hx95_nh112}$, and variable fertilizer application intensity (fertilizer application per unit of land) through $\frac{nh6_3}{hx95_nh112}$. (1 point)

2. What sign do you predict β_1 will have? Why? (1 point)

3. Run the regression and report the estimated coefficients, their standard errors, and R^2. Does the estimated value of β_1 agree with your prediction? (2 point)

4. Interpret the value of the estimated parameters $\hat{\beta}_0$ and $\hat{\beta}_1$ from the model. (2 point)

5. Compute the fitted value $\widehat{yield}_i$ and the residual $\hat{\mu}_i$ for each observation, and verify that the residuals (approximately) sum to 0. (1 point)

6. According to the estimated relation, what is the predicted $\widehat{yield}_i$ for a farmer with a fertilizer application intensity of 100 kg/mu? (pay attention to units). (1 point)

7. How much of the variation in yields for these farmers is explained by their fertilizer input intensity? Can you provide some reasons for why yields might be lower among poorer farmers? (2 points)

8. An professor asks you whether they should estimate a linear relationship between yields and fertilizer input. You are thinking of introducing quadratic form in the regression. Explain why are you interested in quadratic regression form and report your results and interpretations (3 points)

9. The professor sees your analysis. She asks you to explore alternative functional forms for this relationship. Using the provided data, you estimate the following equations:
   * A log-linear relationship: $log(yield_i) = \beta_0 + \beta_1\, fertilizer_i + \mu_i$
   * A linear-log relationship: $yield_i = \beta_0 + \beta_1\, log(fertilizer)_i + \mu_i$
   * A log-log relationship: $log(yield)_i = \beta_0 + \beta_1\, log(fertilizer)_i + \mu_i$

   `We take natural log here. In Stata, the coding is "log(variable+1)"`

   Run the regressions recommended by the professor. Interpret the parameters on fertilizer application intensity (or log(fertilizer application intensity)) in each of these equations. (3 points)

10. Repeat *question 1* for households in year 2001 only (linear-linear regression). How do your estimates of $\widehat{\beta}_0$ and $\widehat{\beta}_1$ differ from *question 1*? Briefly explain the possible reasons in 3-5 sentences. (2 points)

11. Describe the SLR4 assumption under this case. Do you think that assumption holds true? (1 point)

Question 2

A large-landholding farmer outsmarts others by running an experiment to measure the effect of fertilizer application on agricultural yields. She **randomly assigns** her 400 land plots into two groups. In the first group, she inputs 20 kg/mu of fertilizers, while the other group receives 40 kg/mu of fertilizers. She also keeps other agricultural inputs (water, pesticides, labors) equally between these two groups. Let Y_i denote the yields for i^{th} plot, let X_i denote the amount of fertilizer application per unit of land, ($X_i = 20$ or 40), and consider the regression model $Y_i = \beta_0 + \beta_1 X_i + \mu_i$

1. Explain what the term u_i represents. Proposal a potential reason that why different plots might have different values of u_i? (2 point)

2. Is $E[u_i|X_i] = 0$ in this case? Explain why or why not? (1 point)

3. The estimated regression is $Y_i = 300 + 0.7X_i + \mu_i$.

   (a) Is $\beta_1 = 0.7$ an unbiased estimator? (1 point)

(b) After you got this estimation, the farmer's assistant tells you that he is concerned that the program may not actually have randomly allocated fertilizer quantity across the 400 plots, and that some cheating may have gone on (he heard that the richest land (land with highest soil quality) were more likely to be put into the more fertilizer group). What concern would this give rise to in your estimation? (1 point)

(c) What additional variable (data) would you like to collect to verify whether the assistant's concern is true, and what regression specification would you use these data for to investigate whether his concern is true? (3 points)